

CaseMaster: Designing a Probe for Oral Case Presentation Training with LLM Assistance

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Abstract

Preparing an oral case presentation (OCP) is a crucial skill for medical students, requiring clear communication of patient information, clinical findings, and treatment plans. However, inconsistent student participation and limited guidance can make this task challenging. While Large language models (LLMs) can provide structured content to streamline the process, their role in enhancing these skills and integrating into medical education remains under-explored. To address this, we conducted a formative study with six medical educators and developed *CaseMaster*, an interactive probe that leverages LLM-generated content tailored to medical education to help users train their OCP skills. Through a preliminary user study from the expert perspective, we validated the effectiveness of the probe.

CCS Concepts

• Human-centered computing → User interface toolkits.

Keywords

Large language models, medical education

ACM Reference Format:

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1 INTRODUCTION

The oral case presentation (OCP) is a fundamental mode of physician-to-physician communication [14], serving as a structured format for

conveying patient information, clinical findings, diagnostic reasoning, and therapeutic plans [39]. Its importance as an educational goal for learners has been recognized by the Clerkship Directors in Internal Medicine (CDIM) [13] and many other medical educators [2, 12]. A well-executed case presentation enhances the clarity and efficiency of information transfer between physicians, ensuring that critical patient details are accurately communicated.

To achieve proficiency in OCP, medical educators emphasize the necessity of robust training methods, employing techniques such as case discussions [42], case writing [43], and role-playing [37] individually or in combination to maximize learning outcomes and practical skills. Students participate in case discussions by analyzing clinical cases with a facilitator, draft and review detailed reports in case writing before oral presentations, and simulate clinical scenarios through role-playing exercises, acting as either doctors or patients. These approaches collectively enhance students' OCP abilities effectively [30].

However, several challenges can hinder the effectiveness of the above-mentioned methods. First, while these approaches are beneficial for developing critical thinking and communication skills, they can sometimes be dominated by more vocal students, leaving quieter individuals less engaged [31, 54]. This imbalance can create a learning environment where not all students benefit equally. Although techniques like Round-Robin Participation [10] and the Six Thinking Hats method [60] can help increase student engagement, some students still lack sufficient training. Additionally, while numerous studies have developed training systems that provide automated feedback on verbal communication [52, 53, 55], these systems primarily focus on public speaking and often require users to specify parameters such as spoken emphasis or voice modulation. Although these aspects are important, they do not fully address the specific challenges of OCPs. Moreover, automated feedback systems for verbal communication [52, 53, 55] focus mainly on public speaking and overlook the unique challenges of OCPs. Additionally, students struggle with unclear expectations for OCPs, making it hard for them to gauge their performance without defined goals and feedback. This uncertainty can hinder training and lower confidence. Lastly, implementing these methods requires significant time and resources [15], making it difficult to balance with other

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curriculum demands [6], and instructors may struggle to provide sufficient hands-on training [19].

Numerous studies have explored the use of large language models (LLMs) in education, highlighting their potential for personalized, efficient, and engaging learning experiences [1, 3, 28, 33, 45]. These advancements offer promising solutions for developing OCP skills by providing adaptive, in-situ-generated content aligned with educators' insights [34, 35]. However, without structured integration, LLMs can lead to fragmented training, with outputs lacking organization or direct relevance to educational goals. While some studies [40, 59] combine LLMs with Knowledge Graphs (KG) to improve accuracy and structure, these methods may struggle with complex, context-dependent medical scenarios [21, 62]. Additionally, gaps remain in understanding the design requirements for such training and how medical practitioners will perceive and engage with these technologies.

In response, we examined the effectiveness of structured training frameworks in helping medical practitioners learn OCPs and identified any additional support needed. We collaborated with six educators to evaluate educational interventions aimed at improving students' OCP skills. As part of this collaborative effort, we developed a system probe named *CaseMaster* to assist medical students in training their OCP skills. The probe guides users through preparation and reflection stages, enabling them to explore patient cases, prepare using the SOAP format with LLM-generated content, and reflect by comparing their solutions with reference answers, integrating feedback to enhance their skills and confidence in a structured learning experience. Finally, through a preliminary expert evaluation, we identified the limitations of the current probe and explored potential improvements to enhance its effectiveness in future iterations.

2 BACKGROUND and RELATED WORK

2.1 Oral Case Presentation in Medicine

The oral case presentation (OCP) is a crucial communication skill in medicine [14, 39]. Studies have examined educator expectations for OCP within the field [14, 36]. For instance, Green et al. [14] conducted a significant survey of 110 members of the CDIM to understand common expectations for OCPs among clinical clerks. Other research has explored various methods to enhance OCP skills. Kim et al. [23] conducted a randomized controlled study that demonstrated how using encounter cards, which provided structured and timely feedback, effectively improved OCP presentation skills among medical students. Chan et al. [7] reviewed medical educators' expectations for effective case presentations, highlighted the absence of a comprehensive model for teaching OCPs, and advocated for a performance-based approach to enhance structured feedback and teaching support for medical students. Simultaneously, parallel research efforts focus on developing frameworks and curricula to improve OCP skills among medical students [9, 18]. For example, Williams et al. [57] created a curriculum to improve medical students' OCP skills through instruction, self-evaluation, and peer evaluation, aiming to enhance quality while reducing faculty time. Wawrykow et al. [56] developed a competency-based tool and curriculum for OCPs in emergency medicine. These studies

highlight the ongoing efforts to standardize and improve educational strategies for training OCP skills. The alignment in educator expectations suggests that a unified approach to training and evaluating OCPs could be highly beneficial. In this study, we collaborate with six medical educators to explore the use of LLMs for creating probes tailored to OCP skill development.

2.2 Training Systems for Enhancing Presentation Skills

Given the strong similarities between OCP skills and general presentation training—both emphasizing clear communication, effective delivery, and audience engagement—we reviewed existing studies in training systems to inform our approach. Various systems in the HCI field provide automated feedback on presentation quality by analyzing vocal features such as pitch, speech rate, and loudness [25, 38]. However, many existing systems rely on fixed thresholds for assessment, which often lack the contextual nuance needed for targeted improvements [44]. In addition, the feedback strategies used can significantly influence training outcomes. While real-time feedback offers immediate insights that help users correct mistakes quickly [48, 52], research also suggests that intermittent feedback plays a crucial role in enhancing long-term retention and fostering sustained learning and skill mastery [27, 46, 47]. In the OCP training scenario, the focus is on self-reflection and iterative improvement, requiring users to integrate reflective thinking with interactive engagement. Drawing on insights from previous work, our approach prioritizes effective content delivery in OCP skill training, emphasizing practical, problem-oriented presentations. Unlike traditional methods that focus on vocal metrics like tone and speed to engage the audience, we emphasize content structure and coherence to ensure logical organization and clarity for more effective communication.

2.3 Large Language Models in Education

Recent advancements have shifted educational applications towards LLMs, which provide personalized, efficient, and engaging learning experiences [33]. LLMs can generate tailored self-study materials, like flashcards and course-specific questions, to meet individual needs [5, 8, 41]. For instance, Divito et al. [11] used ChatGPT to assist medical learners in problem-based learning by clarifying complex queries [24]. Huber et al. [20] introduced game-based methods to enhance skill development while reducing over-reliance. On the other hand, LLMs provide educators with powerful tools to streamline assessments and deliver personalized feedback [4, 58]. For example, Mitra et al. [32] developed *RetLLM-E* to help educators address frequently asked questions by synthesizing responses from previous discussions and course materials. Further, recent approaches view LLMs as a powerful backend for agent-based simulations [26, 50, 51], presenting a promising avenue for exploration. However, there is limited research on how medical practitioners can utilize LLMs to optimize workflows in medical education. Traditional education methods for OCP training rely heavily on practical experience, such as case discussions [42] and role-playing [37]. The integration of LLMs offers opportunities to streamline training and workflows; however, it also raises concerns about their reliability and adaptability in education.

3 FORMATIVE STUDY AND DESIGN GOALS FORMULATION

To design the probe, we conducted a formative study with six medical educators. The participants, with an average age of 29.83 (SD = 5.15), included three females and three males, identified as P1 through P6. Half of the participants (P2, P5, P6) are novice educators with less than three years of teaching experience, while the others (P1, P3, P4) have over five years of experience guiding junior students. All had prior experience in medical education, ensuring their expertise was aligned with our goal of enhancing OCP skills.

Procedure. The participants engaged in semi-structured interviews lasting approximately 45 minutes. The discussions aimed to uncover the methods and practices employed in the routine preparation of OCPs. Questions explored various aspects, such as whether participant follow a specific templates when organizing case information (e.g., “Do you have a specific structure you follow when preparing case information?”), the types of resources they consult (e.g., “Which materials do you find most useful—textbooks, specialized monographs, or online databases?”), and how they integrate educational theories into their presentations (e.g., “How do you apply theories like SOAP [49] in your presentations?”). The challenges encountered during the preparation process were also examined to identify obstacles that may hinder effective OCP (e.g., “What are the most significant challenges you face in preparing for an oral presentation?”). Also, we explored their perspectives on utilizing LLMs, such as ChatGPT, in the design of comprehensive OCPs and how they conceptualize LLMs.

Data analysis. The entire process was video-recorded and transcribed for thorough analysis. We employed affinity diagrams [17, 29] to systematically categorize and organize the data. To minimize subjectivity, the research team held weekly meetings to discuss and reach a consensus on key findings. We asked participants in later sessions for their views on findings from earlier ones, enriching our understanding and helping to validate the results. Cross-referencing with the Miro board further ensured a rigorous, iterative analysis, offering deeper insights into participant profiles and the challenges they encounter in practice. This informed the interpretation of their responses.

After completing the analysis, six participants reviewed the results to assess alignment with their experiences, providing feedback to refine our findings for greater accuracy. They were asked to annotate and provide feedback on aspects that either reflected or contradicted their experiences. This feedback allowed us to refine our analysis, ensuring it more accurately represented the participants’ perspectives. Finally, we consulted two external medical educators to identify potential blind spots and ensure our findings aligned with current medical education practices.

Findings. As illustrated in Figure 1, this represents a typical example of an oral case presentation. All participants emphasized the importance of comprehensive preparation for OCPs. The process typically begins with retrieving guidelines and summarizing key points. The time invested in preparation varies depending on the complexity of the case and the participant’s experience, with seasoned educators generally spending less time than novices. During preparation, they refer to a range of resources, including

textbooks, specialized manuals, and consensus guidelines, while literature is consulted less frequently. To ensure the case information is well-organized, participants consistently use specific templates or structures. All participants indicated that they primarily prepare for OCPs using notes, typically in Word documents or PowerPoint slides. P2 and P4 mentioned that they occasionally take screenshot from textbook and annotate the text to remind themselves of additional information, such as explanations or supplemental examples. They unanimously agreed that having a document outlining the main flow and content of the presentation was beneficial, as it allowed them to “easily conduct the presentation based on the documented plan” (P6).

Based on the feedback, we identified several challenges in traditional OCP training and established six corresponding design goals for the probe to address these issues. First, educators pointed to the insufficient support for novice presenters, particularly in applying frameworks like SOAP. To address this, the probe should provide adaptive guidance, assisting learners in applying frameworks flexibly across different case scenarios (DG1). Second, common mistakes such as a lack of focus and failure to refer to notes were noted. The tool should offer targeted guidance to help learners recognize and correct these errors, improving presentation skills (DG2). Third, educators emphasized the need for timely and constructive feedback, as indirect or delayed feedback diminishes learning effectiveness. The probe must deliver real-time, actionable feedback to enhance skill development (DG3). Furthermore, while educators acknowledge the potential of LLMs like ChatGPT, concerns about their integration remain. Therefore, the tool should be intuitively designed to integrate LLMs into OCP training without requiring deep technical knowledge (DG4). Additionally, educators noted the cognitive load associated with crafting effective prompts for LLMs. The probe should mitigate this by providing pre-set prompts, thereby reducing the cognitive effort for learners (DG5). Finally, concerns about the reliability and depth of LLM outputs in specialized fields call for flexible controls to refine outputs and ensure accuracy (DG6). These design goals aim to create a more effective and supportive learning environment for both learners and educators.

4 PROBE: CASEMASTER

Following methodologies outlined in several medical education studies [9, 18], we collected data for *CaseMaster* primarily from educational scenarios and select instructional videos. This approach, which includes recordings of simulated patient presentations and course examples, is grounded in proven educational practices. The scenarios are designed to closely mirror real-world clinical settings, providing students with realistic and challenging cases to enhance their diagnostic and communication skills. In total, we obtained 30 oral case presentations, each accompanied by relevant patient medical records. These records were processed and organized into five key groups: *Patient Information* (Name, Gender, Chief Complaint), *History of Present Illness* (HPI), *Past Medical History* (PMH), *Physical Examination*, and *Laboratory Tests and Imaging Studies*. Each case is also paired with a corresponding reference presentation answer.

The quality of the data has been validated by six medical educators. To facilitate student learning, each case’s difficulty is categorized and scored into three levels: simple, intermediate, and

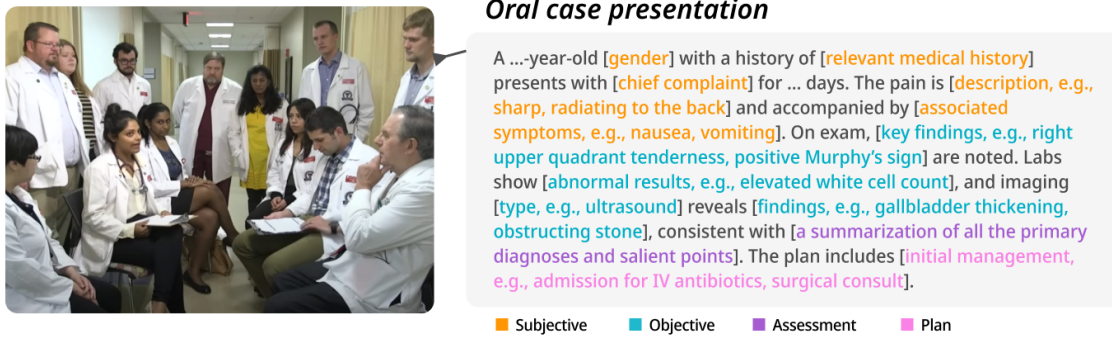


Figure 1: Example of an oral case presentation: On the left, a group of healthcare professionals, including doctors and medical students, are gathering in a clinical hallway for a case presentation, with the presenter leading the discussion while the others listen attentively. On the right, the case presentation is structured into four key sections: Orange (Subjective) for history and symptoms, Blue (Objective) for findings, Purple (Assessment) for diagnosis, and Pink (Plan) for treatment.

advanced. We obtained IRB approval from the Research Ethics Committee, and all data have been anonymized using pseudonyms.

System Overview. We meticulously designed *CaseMaster*'s interface and interactions based on design goals identified through the formative study. The probe guides users through two key stages: preparation and reflection. In the preparation stage, users start by viewing a patient list, organized from simple to complex cases (DG1). As they explore each patient's details, they interactively access resources and prepare their case using the SOAP format. *CaseMaster* offers tailored materials and strategies for each step, with pre-set activities aligned with typical queries for the LLM (DG5). Users can regenerate content, edit drafts, and review their interaction history (DG2). After completing their preparation, users record and upload their solutions for assessment. In the reflection stage, users transcribe their solutions, compare them with a reference answer, and receive a score based on a combination of an educator-provided score sheet and LLM feedback (DG3). This stage fosters reflection, feedback integration, and further learning to enhance their skills. By combining these stages with adaptable user controls, users are empowered to navigate LLM-generated content with ease and intuitive design (DG4), creating a dependable and confidence-inspiring experience throughout their training (DG6). We also provide a System walkthrough in Appendix C.

Preparation Stage. In the Preparation Stage, *CaseMaster* is organized into interactive components designed to streamline the creation and refinement of medical case presentations. The process begins with the *Patient List* (Figure 2-A), a structured overview where users can browse through patient records from the Department of Orthopedics. Each entry provides essential details, such as the patient's name, age, and a complexity indicator, serving as a quick reference for selecting a case. The table also features a search function, allowing users to quickly locate specific patients.

Upon selecting a patient, users are directed to the *Patient Record* (Figure 2-B). This section offers a comprehensive summary of the patient's medical history and current status, organized into distinct subsections. The *Identifying Information* block includes demographic details, while the *History of Present Illness (HPI)* provides a detailed narrative of the patient's symptoms, progression, and

underlying conditions. The *Laboratory Test* block lists diagnostic results, with abnormal values prominently highlighted for easy identification. Additionally, the *Medication Use* block outlines the patient's current treatment regimen, including dosages, frequencies, and administration notes. This structured layout ensures that all relevant information is readily accessible for thorough case analysis.

The main focus of the case preparation process occurs in the *Preparation Panel* (Figure 2-C), where users draft their case presentations. This panel is structured around the SOAP framework, allowing users to customize their notes according to their specific needs. If a user deems a section, such as the *Objective*, to be complete, they can mark it by clicking the corresponding checkbox (Figure 2-C1). These checkboxes serve as visual cues, helping users structure their drafts effectively.

During each draft revision, users can click the "Open the assistant" button to activate the LLM assistant, which aids in refining their drafts (Figure 2-C2). The assistant comprises the *Action Set*, *Input Area*, and *Output Area*. The *Action Set* includes a range of pre-set buttons that perform specific tasks, such as evaluating content, conducting content searches, and providing presentation advice. Appendix A lists the activities implemented in our probe, based on findings from the formative study. Users can also expand their ideas by creating custom activities using the LLM below the pre-set buttons. The *Output Area* features a Markdown editor that displays the LLM's outputs after users interact with the *Action Set* or make custom queries. Users can select content from the LLM's outputs and click the "Copy" button at the bottom of the assistant panel to copy it to the clipboard, allowing them to paste it anywhere within the current section's editor. The "Stop Generating" button halts the LLM's output, while the "Close Assistant" button hides the assistant. The "History Record" button provides access to previously generated content.

Users can draft key points for their final presentation and simulate the scenario by conducting an oral presentation by clicking the "Voice" button (Figure 2-C3), much like TOEFL/IELTS practice [16, 61]. Although real-life simulations with actual partners provide a more immersive experience, this method is more practical.

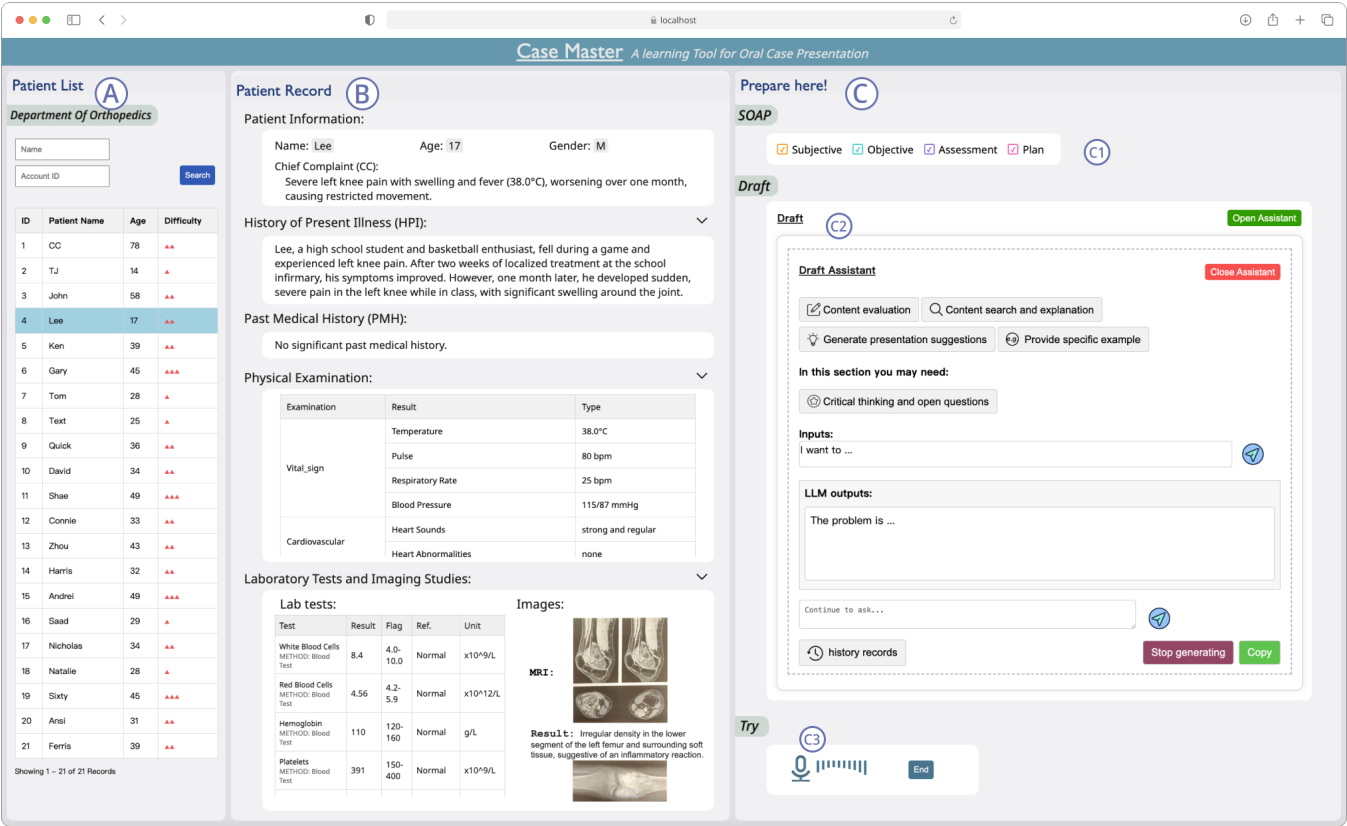


Figure 2: During the Preparation Stage, users can choose a patient from the (A) Patient List, review detailed information in the (B) Patient Record, and draft and refine their presentation within the (C) Preparation Panel. The Preparation Panel includes LLM-powered assistance for content evaluation, suggestions, and generating specific examples to support users in crafting their presentation.

It offers flexible, on-demand practice and allows users to focus on specific areas at their own pace.

Reflection Stage. Upon completing the presentation, the probe transitions into the Reflection Stage, where *CaseMaster* meticulously assesses the quality and thoroughness of the users' presentations. This stage consists of two main components: the *Validation* (Figure 3-D) and the *Score Sheet* (Figure 3-E). In the Validation section, users transcribe their responses and compare them to a reference answer. Discrepancies or missing elements are automatically highlighted, with distinct background colors indicating areas that need further review and understanding. Users can click on the highlighted sentence to view a detailed explanation of the identified mistakes.

In traditional educational settings, instructors often rely on detailed score sheets to evaluate students' video or audio submissions. To enhance efficiency and consistency, *CaseMaster* leverages LLMs to automatically generate scores based on a comprehensive *Score Sheet* and a reference answer. The LLM compares the student's work against the reference, identifying discrepancies and assessing the accuracy and completeness of each section.

In the *History* section, the LLM assesses the timing and characterization of symptoms, as well as the inclusion of pertinent facts,

using a 0 to 3 scale. The *Important Information* section evaluates the relevance and focus of the medical history, ensuring unnecessary details are avoided, with scoring criteria that include both numerical scores and Yes/No evaluations. For the *Physical Examination*, the LLM checks if vital signs are prioritized and whether the examination data is relevant to the diagnosis, again scoring from 0 to 3.

The *Labs* section measures the inclusion and understanding of essential lab data, while *Assessment and Plan* scores the clarity of the synthesis statement and the logic of differential diagnoses. The *General and Style* section evaluates the overall organization, timing, and delivery of the presentation, scoring elements like duration, avoidance of filler words, and the strength of the argument on a scale from 1 (Beginner, Painful) to 3 (Masterful, Inspirational).

To ensure thorough and objective grading, the LLM integrates these criteria with the reference answer, delivering precise feedback that identifies areas where students meet expectations and where improvement is needed. On the *Score Sheet*, scores in each category are highlighted in red with a checkmark-✓, mimicking the real-world grading style of an educator. Users can click the checkmark to access detailed explanations for each score as provided by the LLM, including any detected mistakes.

Validation (D)

Solution

Lee is a 17-year-old male with a significant history of recent trauma to the left knee, initially managed conservatively with temporary improvement. However, the subsequent exacerbation of symptoms, including severe pain and swelling, prompted further evaluation. Clinical examination corroborated the presence of inflammation, as indicated by the elevated white blood cell count, and imaging studies confirmed a pathological process within the lower femur. PET-CT was particularly revealing, showing characteristic signs of a high-grade osteosarcoma. The patient's management plan involves amputation, aimed at local control and chemotherapy to address any residual disease. The multidisciplinary team has discussed the case, and the patient and family are fully informed of the treatment strategy and potential benefits and limitations of prosthetic rehabilitation.

Reference

Lee is a 17-year-old male high school student who is an avid basketball player. He presented with a history of falling during a game, resulting in pain in his left knee. Initial treatment in the school infirmary provided temporary relief, but he experienced a recurrence of severe pain and swelling in the left knee about a month later. His vital signs were notable for a temperature of 38.0°C, pulse of 80 bpm, respiratory rate of 25 bpm, and blood pressure of 115/87 mmHg. Examination revealed marked swelling and tenderness in the left knee, with limited range of motion. Laboratory tests indicated elevated white blood cell count, normal red blood cells, hemoglobin, and platelets. His renal function and liver enzymes were within normal limits, except for elevated alkaline phosphatase and creatinine. Tumor markers showed a slightly elevated ferritin level. Imaging studies, including X-ray and PET-CT, revealed irregular density in the lower segment of the left femur and abnormal morphology, suggestive of a high-grade osteosarcoma. He has been scheduled for surgery, including amputation, followed by adjuvant chemotherapy to improve the prognosis.

Score Sheet (E)

	Fails To Include	Novice	Competent	Master
	0	1	2	3
History				
Timing and characterization of symptoms	0	1	2✓	3
Includes pertinent facts but excludes extraneous information needed to establish and modify a differential	0	1✓	2	3
Important Information				
Relevance and focused reporting of medical history, family history, surgical history, allergies, medications, and social history	0	1		
Avoids a separate review of system				
Physical Examination				
Vital signs first			No✓ Yes	
Focused physical examination relevant to the diagnosis, includes data necessary for the differential diagnosis but excludes extraneous information	0	1	2✓	3
Labs				
Includes laboratory data essential to the diagnosis and excludes irrelevant data	0✓	1	2	3
Demonstrates understanding of which labs are relevant and which are not	0	1✓	2	3
Assessment and Plan				
Synthesis statement	0	1	2	3✓
Assessment includes a list of at least three differential diagnoses with arguments for and against each. Arguments are succinct.	0	1	2✓	3
General and Style				
		Beginner, Painful	Competent, Not Painful	Masterful, Inspirational
Duration	1	2	3✓	
Organization in the proper order	1✓	2	3	
Use of distractors (uhs, uhms, ahs)	1	2	3✓	
Makes a case	1	2	3✓	

Figure 3: During the Reflection Stage, users can compare their solutions in the (D) Validation panel, where highlighted discrepancies emphasize the differences between their solutions and the reference. The (E) Score Sheet displays LLM-generated scores across various evaluation criteria, with clickable red checkmarks offering detailed feedback.

Following the validation and assessment of the presentations, users can click the “Return” button to navigate back to the Preparation Stage, where they can either refine their existing presentations or select a new patient case to initiate a new round of presentations.

5 PRELIMINARY USER STUDY

We conducted a user study to evaluate our probe from the expert perspective, involving think-aloud sessions and interviews with five experienced medical educators. Their insights highlighted opportunities to better align the tool with broader educational goals and revealed nuances that students might not be aware of, such as the appropriateness of LLM support and the overall effectiveness of the feedback mechanisms. The preliminary user study engaged five experienced medical educators who evaluated *CaseMaster*, providing insights into how they interact with and perceive the tool.

Participants. The group of five educators (2 females, 3 males) comprises three with 2 to 3 years of teaching experience and two with 5 years of experience. Educators E1-E3 were the same individuals as participants P1, P3, and P4 from the formative study, while E4 and E5 were newly introduced to the study.

Tasks. The total duration of each expert’s participation was 45 minutes. Each expert’s session began with an introduction to the research background and objectives. For E4 and E5, we provided an example of an OCP and SOAP format notes to ensure a solid understanding of the concepts. This was followed by a 10-minute tutorial on using *CaseMaster*. The experts were then given 20 minutes to complete tasks involving two patient cases while thinking aloud. The session concluded with a 15-minute semi-structured interview, guided by the questions outlined in Appendix B.

Data collection and analysis. The qualitative data comprises two components: insights gathered during system use and responses

from semi-structured interviews conducted afterward. All data were transcribed, and for the expert evaluation, two authors independently reviewed the transcripts to extract key insights, which were then synthesized into affinity diagrams. Affinity diagrams were preferred over grounded theory due to their widespread use in HCI and interaction design [17, 29]. After this process by the authors, the experts were individually provided with the affinity diagrams and asked to annotate and provide feedback on aspects they found either reflective of or inconsistent with their intentions and experiences. Finally, they engaged in discussions to reach consensus and refine the final findings.

Feedback. Experts generally recognized the value of *CaseMaster* in enhancing case presentations and reducing educators' workload. They appreciated the system's two-stage design, which facilitated students' preparation and reflection, enabling quicker feedback and more focused training. While the reflection stage was praised for reducing teaching burden and offering constructive feedback, some experts were cautious about the LLM's reliability, particularly in areas like term definitions, where discrepancies with students' current knowledge could arise. Despite these concerns, the system was seen as helpful in streamlining thought processes and supporting students' critical thinking skills during presentations. However, experts suggested improvements, including customizable prompts and logic flow visualizations, to further enhance usability and training effectiveness. Additionally, they emphasized the potential of the LLM's solution evaluation feature for providing detailed, real-time feedback to students. Overall, *CaseMaster* was regarded as a valuable tool for supporting medical education, with room for refinement to better meet students' training needs.

6 DISCUSSION AND FUTURE WORK

In developing *CaseMaster*, we aimed to enhance medical education by integrating LLMs to support active training and critical thinking. We designed structured activities that prompt students to engage with content and refine their problem-solving skills, with LLM-generated feedback guiding reflection and revision. To ensure reliability, we grounded the system's content in trusted educational resources, avoiding overly complex information. We also explored the need for flexible prompt customization, allowing educators to tailor instructions to students' needs. In broader terms, we highlighted the value of integrating multi-modal training, such as visual tools and simulations, to accommodate diverse learning styles. All experts assert that evidence-based critical thinking is indispensable to this training process, relevant to both students and LLM logic. While LLM responses may not always be consistent, our focus is on ensuring they have clear, reliable sources and logical reasoning, aligning with educational goals and fostering a symbiotic relationship between students and the system [22]. LLMs can aid advanced tasks, early-stage medical education should prioritize trusted resources, using LLMs as supplementary tools. Finally, combining LLMs with traditional instructor-student interaction in a hybrid model could enhance engagement and improve training outcomes.

Our work has several limitations that warrant further exploration. First, the current patient data, collected by experts, focuses solely on orthopedic cases. Expanding the probe to support training cross various medical departments will require contributions

from a broader range of experts or integration with online knowledge bases. Second, although experts acknowledge *CaseMaster* as a valuable tool for supporting student training, future work should involve conducting a controlled study targeting students that compares *CaseMaster* with a baseline system to assess its effectiveness. Finally, to promote broader adoption, future iterations of *CaseMaster* could include features that allow users to save content as digital materials for future review, or enable seamless export and sharing in slide format for presentations and collaborative purposes.

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A Proposed LLM-Assisted activities

Activity	Response Example
Search key knowledge points	"Given the patient's symptoms, key indicators to consider include limited range of motion, swelling, and localized pain, which could suggest a potential rotator cuff injury."
Review medical literature	"According to recent guidelines from the American College of Cardiology, initiating statin therapy in patients with LDL levels above 190 mg/dL is strongly recommended to reduce the risk of cardiovascular events."
Check the logic of the content	"The treatment plan you outlined is logical, prioritizing conservative management such as physical therapy and NSAIDs before considering surgical options, which aligns with best practices for managing osteoarthritis of the knee."
Assess the reasonableness of content	"Given the patient's active lifestyle and the severity of the ACL tear, the recommendation for surgical reconstruction followed by an aggressive rehabilitation program appears reasonable. Continued monitoring during recovery will be essential."
Provide definitions of terms	"A comminuted fracture refers to a type of bone fracture where the bone is broken into multiple pieces, often requiring surgical intervention for proper alignment and healing."
Provide specific example	"For instance, 'The patient underwent open reduction and internal fixation (ORIF) to stabilize the distal radius fracture following a fall.'"
Explain examples in detail	"This case illustrates how early diagnosis and treatment of compartment syndrome, including a fasciotomy, can prevent long-term muscle and nerve damage in a patient with a tibial fracture."
Give presentation suggestions	"When presenting this case, start with the patient's mechanism of injury, such as a high-energy fall, and how it led to the complex fracture pattern observed, to immediately capture the audience's attention."

B Eight interview questions for experts

RQ1: Case Presentation Outcome

Q1- Please rate the quality of oral case presentations created with **CaseMaster**, compared to those you produce during your usual routine.

Q2- Can **CaseMaster** help you present the case presentations better, such as with the assistance of LLM?

RQ2: Case Presentation Process

Q3- Can **CaseMaster** assist you in effectively managing the oral case presentation process?

Q4- Does the use of **CaseMaster** create additional cognitive demands, like memory or thought strain, when compared to training for oral case presentations without it?

RQ3: Perceptions towards CaseMaster

Q5- Is **CaseMaster** easy to handle?

Q6- Can you use it smoothly without any technical support?

Q7- Which feature (including preparation stage, reflection stage, pre-set activities, and score sheet) provided by **CaseMaster** helps you the most in conducting your oral case presentation?

Q8- Please provide suggestions for improving **CaseMaster**.

C System walkthrough

We demonstrate the use of *CaseMaster* through a detailed workflow scenario, as illustrated in Figure 4. The scenario involves two users, Mary and Bob: one who performs comprehensive preparation and another who engages in reflective analysis. They start by reviewing the patient list (Figure 4 (a)) and select a specific patient, “Lee” (Figure 4 (b)). Lee is a 17-year-old male experiencing severe left knee pain, swelling, and fever (38.0°C). His symptoms have progressively worsened over the past month, significantly restricting his movement. Lee, a high school student and basketball player, initially injured his knee during a game. After receiving localized treatment at the school infirmary for two weeks, his symptoms improved. However, one month later, he experienced a sudden onset of severe pain and swelling in the knee while in class.

Mary began her preparation (Figure 4 (c)) with a Physical Examination. She observed that the patient had a fever of 38.0°C, although other vital signs were relatively normal. The left knee was swollen, exhibited severe tenderness, and had limited movement, indicating possible joint or bone involvement. She documented these findings and noted that,

“The localized nature of the findings, combined with the fever, raised concerns for a serious pathology within the knee joint or surrounding structures. Infectious processes, such as septic arthritis or osteomyelitis, were possible (Figure 4 (c-1)).”

Next, she checked the Laboratory Tests. The patient’s White Blood Cells (WBC) count was normal at 8.4×10^9 /L. She observed,

“Joint infections can cause fever, swelling, and pain. However, Lee’s relatively normal WBC count and the longer history reduced the likelihood of this (Figure 4 (c-2)).”

She then proceeded to query the draft assistant, using the “Content Evaluation” preset activity. The assistant verified that her reasoning was both logical and scientifically sound.

Upon reviewing the Alkaline Phosphatase (ALP) test results, she observed an elevated level of 971 U/L, which was abnormal. This critical finding redirected her diagnostic focus. Recognizing that elevated ALP is commonly associated with bone-related disorders, particularly bone tumors like osteosarcoma, she sought additional information about the ALP test. Using the “Content Search and Explanation” preset activity, she received a comprehensive explanation from the draft assistant. She concluded that,

“Alkaline Phosphatase (ALP) is elevated at 971 U/L, a key finding. ALP is often elevated in bone disorders, particularly in osteosarcoma or other bone-forming tumors (Figure 4 (c-3)).”

She then reviewed the imaging studies, which revealed irregular density and abnormal morphology in the lower femur. She noted that these findings were consistent with potential bone pathology, reinforcing her concern for a possible bone tumor. She stated,

“These imaging results confirmed my suspicion. Osteosarcoma typically shows destructive bone lesions on imaging, and the diagnosis was now likely confirmed by these findings (Figure 4 (c-4)).”

Finally, she summarized her analysis and developed a treatment plan, concluding that:

“The priority now was to control the disease locally through surgery, followed by chemotherapy to reduce the risk of metastasis. This is the standard approach for high-grade osteosarcoma (Figure 4 (c-5)).”

Bob reviewed his presentation and identified several issues (Figure 4 (e)). Initially, he concentrated on the *Validation* section, addressing each poorly constructed sentence individually.

- Figure 4 (e-1) highlights that,
“The sentence does not adequately describe the specific context of the trauma and the initial treatment in the school infirmary, which are important details in the case.”
- Figure 4 (e-2) shows,
“The sentence erroneously suggests confirmation of a pathological process within the lower femur based on clinical examination findings and lab results, which is not accurate.”

Bob then examined the reference answer to gain a deeper insight into effective presentation skills. He proceeded to review the *Score Sheet*:

- Figure 4 (e-3) notes,
“The presentation included pertinent facts regarding Lee’s history and management but could benefit from excluding some extraneous details that do not directly impact the differential diagnosis.”
- Figure 4 (e-4) points out that,
“The presentation did not include essential lab data in the solution, missing crucial information for the diagnosis.”
- Figure 4 (e-5) observes that,
“The organization could be improved by aligning the flow of information more closely with the logical progression of a case presentation.”

Bob felt he had gained substantial knowledge and was confident in his understanding. He looked forward eagerly to the next study.

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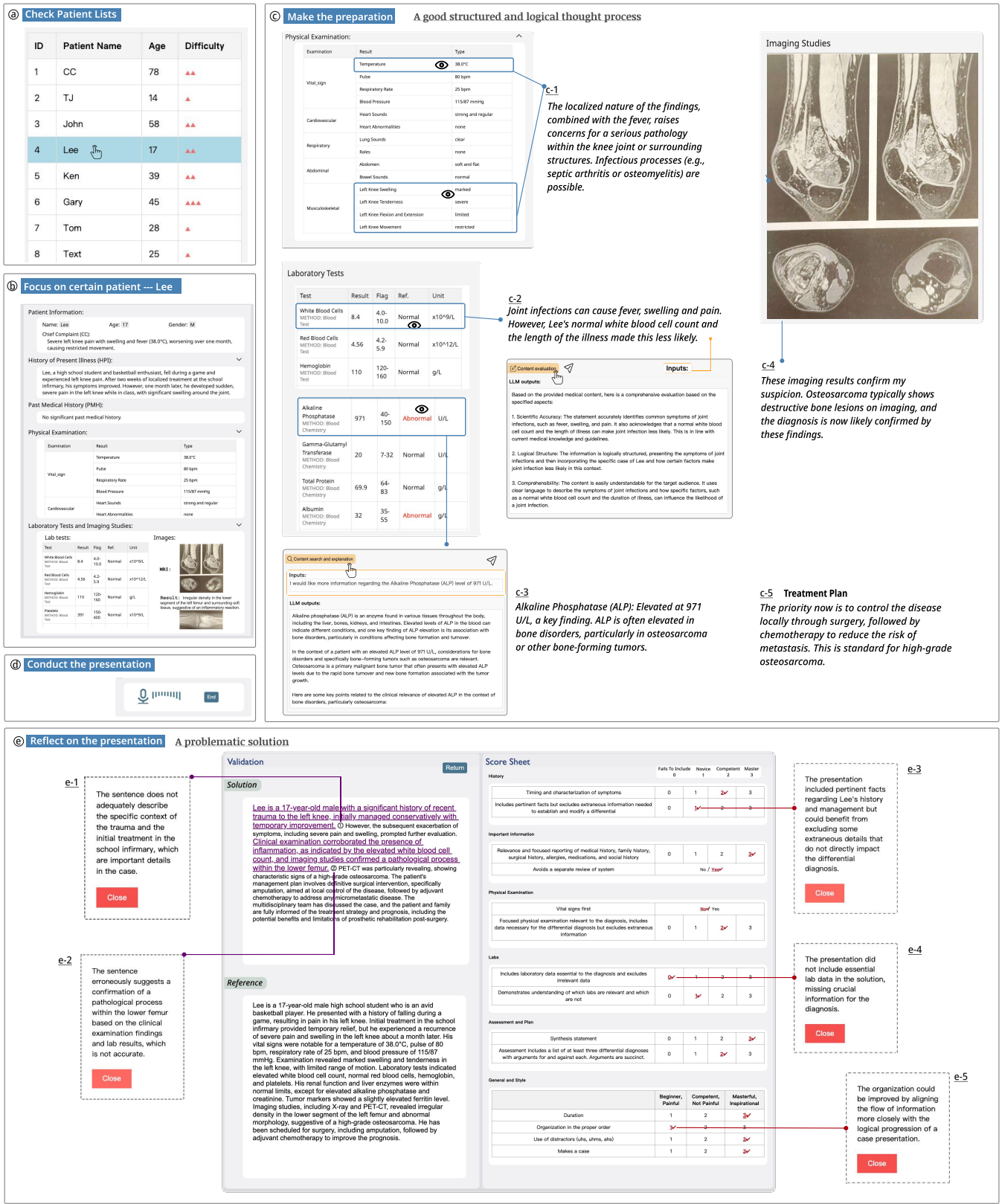


Figure 4: The process of conducting case presentation training with CaseMaster includes the following steps: (a) Reviewing the patient list, (b) Focusing on a specific patient, “Lee”, (c) Demonstrating a well-structured and logical thought process through preset activities and organized preparation, (d) Conducting the oral case presentation, and (e) Highlighting a problematic solution, where another user gets feedback on the presentation.